

LIFE PROCESSES

WORKSHEET - 01

1. Why is diffusion insufficient to meet the oxygen requirements of multicellular organisms like humans?

Solution: Multi-cellular organism's like humans have very big body and require a lot of oxygen to diffuse into body quickly in order to meet the oxygen requirement. Diffusion is a slow process which will take a lot of time to circulate oxygen to all the body cells. Because of its slow nature diffusion is insufficient to meet the oxygen requirements of multicellular organisms like humans.

Q2. What criteria do we use to decide whether something is alive?

Solution:

Walking, breathing, growth and other visible changes can be used to determine whether something is alive or dead. However some living things will have changes that are not visible to our eye; Hence, presence of life process and molecular movements needed to maintain the order in a living structure is a fundamental criteria to decide whether something is alive.

Q3. What are outside raw materials used for by an organism?

Solution: Outside raw material is used by organism for food and oxygen. Raw materials requirement varies on the complexity of the organism and the environment it is living.

Q4. What processes would you consider essential for maintaining life?

Solution: Life processes such as respiration, digestion, excretion, circulation and transportation are essential for maintain life.

Q5. What are the differences between autotrophic nutrition and heterotrophic nutrition?

Solution: Autotrophic Nutrition Organism prepare its own food and is not dependent on any other organism. Food is prepared from CO_2 , water, sunlight. Chlorophyll is required for food preparation Green plants and certain bacteria have autotrophic mode of nutrition.

Heterotrophic Nutrition dependent on other organism for food. Food cannot be prepared from CO_2 , water, sunlight. Chlorophyll is not required for food preparation

Q6. Where do plants get each of the raw materials required for photosynthesis?

Solution: Plants required the following raw material for photosynthesis

1. CO₂ is obtained from atmosphere through stomata
2. Water is absorbed by plant roots from the soil.
3. Sunlight : Plants get sunlight from the sun.
4. Chlorophyll : It is present in chloroplast found in green leaves and green parts of plants